



SUJET STATAPP 332 - 2026-2027

Sujet proposé par : Belpaire Michael - michael.belpaire@sybioticsgroup.com
Organisme : Sybiotics Asset Management
Catégorie : Finance
Sujet :
Constructing Yield Curves in Illiquid Emerging Market

Description du sujet :

In liquid financial markets, yield curves can be estimated from a large number of frequently traded securities. In many emerging and frontier markets, however, government bonds trade infrequently, making standard yield-curve estimation difficult.

The objective of this project is to replicate the methodology developed by Khakimzhanov et al. (2019) in "Constructing a Yield Curve in a Market with Low Liquidity" and to apply it to other illiquid emerging markets.

The paper proposes a Nelson-Siegel yield-curve estimation method adapted to sparse transaction data. Instead of using only same-day observations, the methodology selects recent transactions across different maturity segments and weights them according to their age and transaction volume.

Students will first reproduce the methodology and results for Kazakhstan. They will then apply the same framework to one or several other emerging or frontier markets, for example in Central Asia, the Caucasus, Africa or Latin America, depending on data availability.

The project will also assess the robustness of the methodology by varying the number of observations used, maturity buckets and weighting schemes, and by comparing the estimated curves with alternative specifications such as Nelson-Siegel-Svensson.

The main research question is whether a methodology designed for the illiquid Kazakhstan government bond market can be successfully transferred to other emerging markets.

Données du sujet :

The project will use government bond transaction or price data, including maturity, yield, transaction date and, when available, transaction volume.

Potential sources include central banks, stock exchanges, ministries of finance and financial databases.

Modèles envisagés du sujet :

Nelson-Siegel yield-curve estimation method

Bibliographie du sujet :

Khakimzhanov, S., Mustafin, Y., Kubenbayev, O. and Atabek, D. (2019), "Constructing a Yield Curve in a Market with Low Liquidity," Russian Journal of Money and Finance, 78(4), 71-98.

Nelson, C. R. and Siegel, A. F. (1987), "Parsimonious Modeling of Yield Curves," Journal of Business, 60(4), 473-489.

Notation du sujet :

3

Commentaires :

Sujet adéquat